Double-Angle Formulas and Trigonometric Equations

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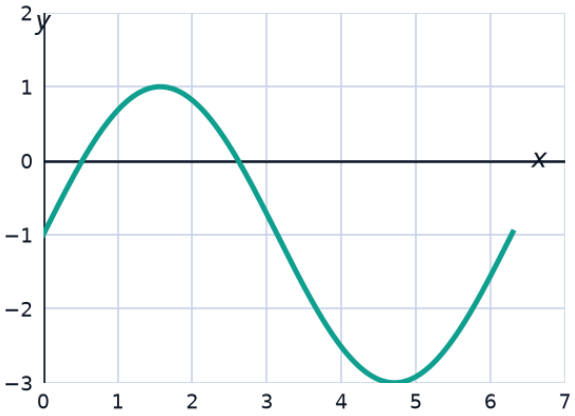
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Double-angle formulas

Use a double-angle formula to write $\cos^2 x$ in terms of $\cos 2x$, and state the corresponding formula for $\sin^2 x$.

2 [Foundational] • Target: Solving trigonometric equations

Solve $2\sin x - 1 = 0$ on the interval $[0, 2\pi)$.



3 [Foundational] • Target: Half-angle formulas

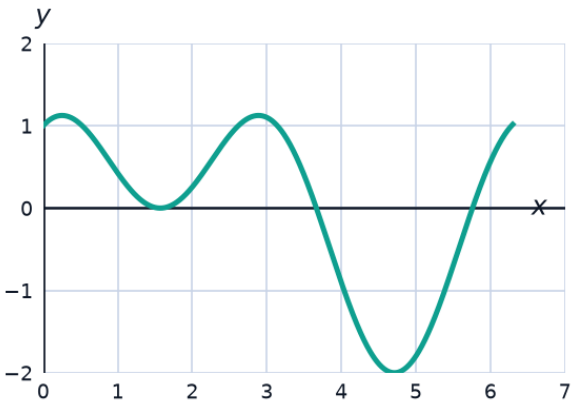
Use a half-angle formula to find the exact value of $\cos 15°$ (using $15° = \frac{30°}{2}$).

4 [Foundational] • Target: Half-angle formulas

Given $\cos\theta = \frac{7}{25}$ with $0 < \theta < \frac{\pi}{2}$, find $\sin\frac{\theta}{2}$ exactly.

5 [Foundational] • Target: Equations requiring double-angle substitution

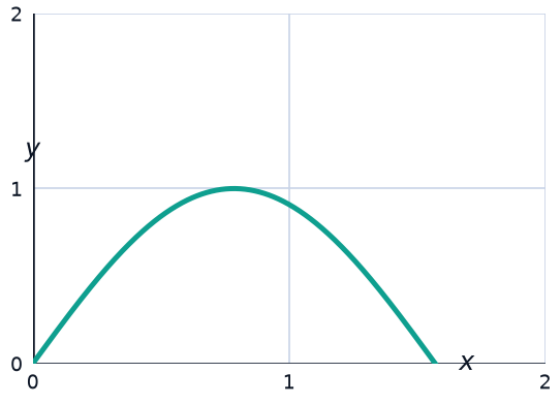
Solve $\cos 2x + \sin x = 0$ on the interval $[0, 2\pi)$, using $\cos 2x = 1 - 2\sin^2 x$.



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6 [Foundational] • Target: Applications of double-angle formulas

A projectile's range is given by $R(\theta) = \frac{v^2 \sin 2\theta}{g}$. Using the double-angle formula, explain why the maximum range occurs at $\theta = 45°$.



7 [Foundational] • Target: Applications of double-angle formulas

Given $\sin\theta = \frac{1}{3}$ with θ in Quadrant I, find $\cos 2\theta$ and $\sin 2\theta$ exactly.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Half-angle formulas for tangent

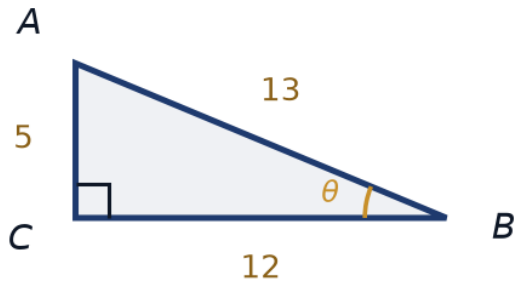
State the half-angle formula for $\tan\frac{\theta}{2}$ in terms of $\sin\theta$ and $\cos\theta$ (the form avoiding a square root), and use it to find $\tan 15°$.

9 [Intermediate] • Target: Equations with multiple angle types

Solve $\cos 2x = 1 - 3\sin x$ on $[0, 2\pi)$, using $\cos 2x = 1 - 2\sin^2 x$.

10 [Intermediate] • Target: Triple-angle preview

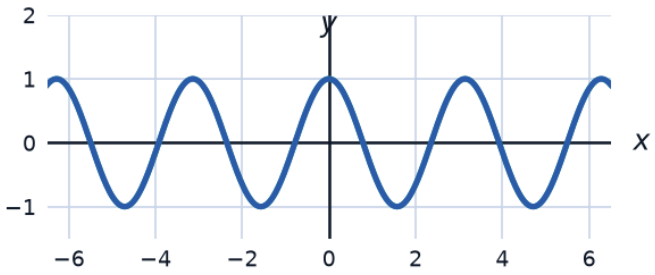
Using $\sin(2x + x) = \sin 2x\cos x + \cos 2x\sin x$ together with the double-angle formulas, derive the triple-angle formula $\sin 3x = 3\sin x - 4\sin^3 x$.



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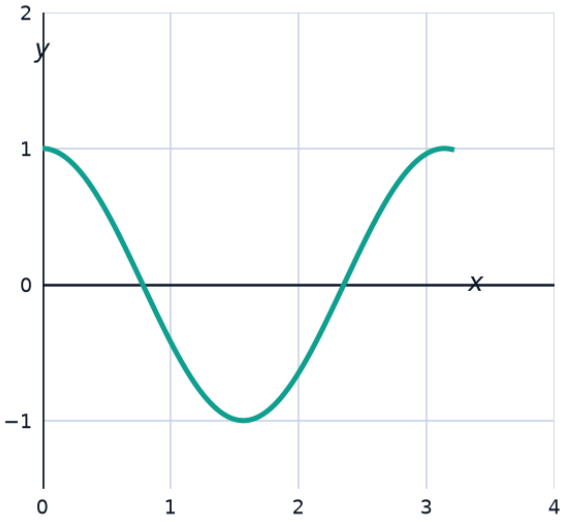
11 [Intermediate] • Target: Graphical verification of a double-angle identity

The functions $y = \cos 2x$ and $y = 1 - 2\sin^2 x$ are graphed together below. Explain what it means that the two curves appear identical, and state the identity this confirms.



12 [Intermediate] • Target: Solving equations restricted to a smaller interval

Solve $2\cos 2x - 1 = 0$ for $x \in [0, \pi)$ only (note the restricted domain).



13 [Intermediate] • Target: Solving trigonometric equations

Solve the logarithmic equation $\log_3(x + 6) + \log_3(x) = 3$ and eliminate any extraneous solutions.

14 [Intermediate] • Target: Half-angle formulas

Write the standard equation of the ellipse having foci at $(\pm 3, 0)$ and vertices at $(\pm 5, 0)$. State the lengths of the major and minor axes, and compute the eccentricity.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

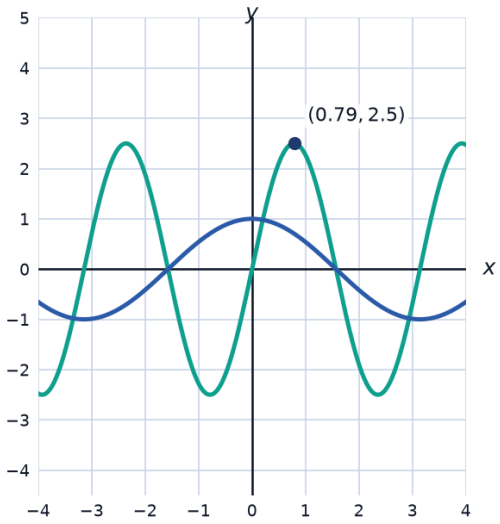
15 [Challenging] • Target: Equations requiring double-angle substitution • Cross-Lesson: 8.03 Introduction to Limits

An ellipse has vertices at $(1, 6)$ and $(1, -4)$, and a focus at $(1, 4)$. Determine its standard equation and compute its eccentricity.

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16 [Challenging] • Target: Applications of double-angle formulas • Cross-Lesson: 9.01 Arithmetic Sequences and Series

Find the vertex, focus, and directrix of the parabola given by the general equation $y^2 + 8x - 4y + 28 = 0$.



17 [Challenging] • Target: More equation practice • Cross-Lesson: 9.02 Geometric Sequences and Series

A radioactive isotope has a decay constant $k = 0.035\ \text{year}^{-1}$. (a) Find its half-life to the nearest tenth of a year. (b) If an initial sample contains $250\ \text{mg}$, how much remains after $40\ \text{years}$?

18 [Challenging] • Target: Half-angle formulas for tangent • Cross-Lesson: 9.03 The Binomial Theorem

[Conic Sections & Analytic Geometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

19 [Challenging] • Target: Equations with multiple angle types • Cross-Lesson: 1.01 Functions and Function Notation

[Exponential, Logarithmic & Logistic Dynamics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

20 [Challenging] • Target: Triple-angle preview • Cross-Lesson: 1.02 Domain and Range

[Polar Coordinates, Complex Numbers & De Moivre's Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Double-angle formulas

Find the half-angle formula for $\cos(\theta/2)$.

a. $\pm\sqrt{\frac{1+\cos\theta}{2}}$

b. $\cos\theta/2$

c. $\pm\sqrt{\frac{1-\cos\theta}{2}}$

d. $\sqrt{1+\cos\theta}$

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22 [Foundational] • Target: Solving trigonometric equations

State the double-angle formula $\sin 2\theta =$

a. $2\cos^2\theta - 1$

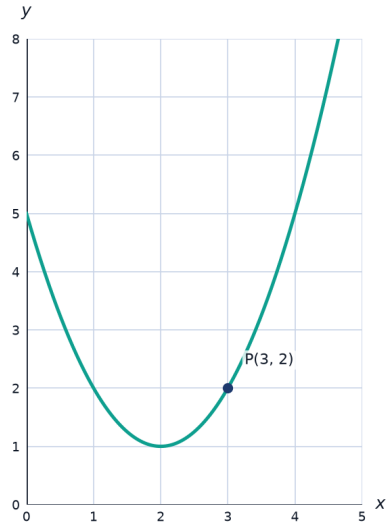
b. $\sin^2\theta - \cos^2\theta$

c. $2\sin\theta\cos\theta$

d. $\cos^2\theta - \sin^2\theta$

23 [Foundational] • Target: Half-angle formulas

The diagram shows part of the curve with equation $y = x^2 - 4x + 5$ and the point $P(3, 2)$ lying on the curve. Find the gradient of the tangent to the curve at the point P.



a. 2

b. 6

c. -4

d. $\frac{2}{3}$

24 [Foundational] • Target: Equations requiring double-angle substitution

A straight line passing through the origin is tangent to both the curve C_1 with equation $y = e^{kx}$, where k is a positive constant, and the curve C_2 with equation $y = \ln x$. Determine the exact distance between the two points of contact.

a. $(e-1)\sqrt{e^2+1}$

b. $e\sqrt{e^2+1}$

c. $(e-1)\sqrt{e^2-1}$

d. $\sqrt{e^4-e^2+1}$

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25 [Foundational] • Target: Applications of double-angle formulas

The curve C has equation $y = \frac{\ln x}{x^2}, \quad x > 0$. The point M is the stationary point of C, and the line L is the tangent to C at M. The finite region R is bounded by the curve C, the tangent line L, and the vertical line $x = 1$. Find the exact area of R.

a. $\frac{4\sqrt{e} - 2e - 1}{2e}$

b. $\frac{4\sqrt{e} - 2e + 1}{2e}$

c. $\frac{2e - 3\sqrt{e}}{2e}$

d. $\frac{2\sqrt{e} - 2e - 1}{2e}$

26 [Foundational] • Target: More equation practice

Consider the system of linear equations represented by the matrix equation $(1, 1, 1; 1, 2, 3; 3, 5, k)(x; y; z) = (6; 14; m)$ where k and m are real constants. Suppose the system is consistent and has infinitely many solutions. Let $P(x, y, z)$ be the unique solution that minimizes the squared Euclidean distance from the origin, $d^2 = x^2 + y^2 + z^2$. What is the value of $k + m + d^2$?

a. 55

b. 47

c. 38

d. 109

27 [Foundational] • Target: Half-angle formulas for tangent

A curve satisfies the differential equation $\frac{dy}{dx} = \frac{k - y}{2x + 1}$ for $x > -\frac{1}{2}$, where k is a real constant. The curve passes through the points $(0, 8)$ and $(4, 2)$. Find the value of k.

a. -1

b. 11

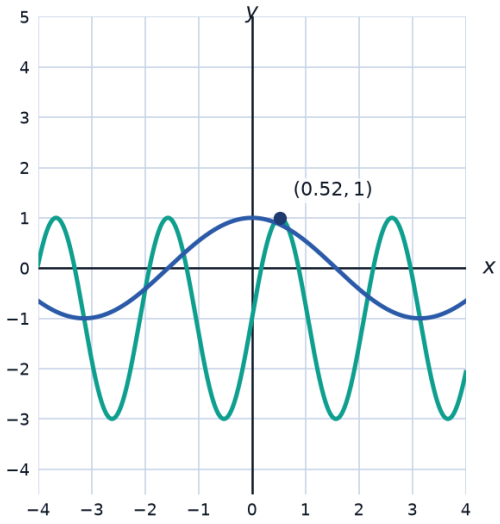
c. $\frac{5}{4}$

d. 5

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28 [Foundational] • Target: Equations with multiple angle types

For a constant k, the equation $\sqrt{2x+k} = x - 1$ has $x = 5$ as a valid solution. What is the extraneous solution produced when solving this equation by squaring both sides?



a. -5

b. -1

c. 1

d. 6

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Triple-angle preview

A curve has equation $y = \ln(2x^3 + 1)$. What is the gradient of the tangent to the curve at the point where $x = 1$?

a. 2

b. 6

c. $\frac{2}{3}$

d. $\ln 3$

30 [Intermediate] • Target: Graphical verification of a double-angle identity

Consider the equation $\sqrt{2x+k} = x - 1$, where k is a constant. If $x = 5$ is a valid solution to the equation, what is the extraneous solution that arises when solving the equation algebraically by squaring both sides?

a. -1

b. 1

c. -5

d. 6

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31 **[Intermediate]** • **Target: Solving equations restricted to a smaller interval**

Find the particular solution of the differential equation $\frac{dy}{dx} = 2xe^{-y}$ given that $y = 0$ when $x = 0$.

a. $y = \ln(x^2 + 1)$

b. $y = \ln(x^2)$

c. $y = \ln(2x^2 + 1)$

d. $y = x^2$

32 **[Intermediate]** • **Target: Double-angle formulas**

State the angle addition formula $\sin(A + B) =$

a. $\cos A \cos B - \sin A \sin B$

b. $\sin A \cos B + \cos A \sin B$

c. $\sin A \cos B - \cos A \sin B$

d. $\sin A + \sin B$

33 **[Intermediate]** • **Target: Solving trigonometric equations**

State the cofunction identity $\sin(90^\circ - \theta) =$

a. $\tan\theta$

b. $\cos\theta$

c. $-\cos\theta$

d. $\sin\theta$

34 **[Intermediate]** • **Target: Half-angle formulas**

Which differential equation correctly models the statement: "The rate of decrease of the temperature T of an object is proportional to the difference between T and the surrounding temperature T_e."?

a. $\frac{dP}{dt} = kP$

b. $\frac{dT}{dt} = -k(T - T_e)$

c. $\frac{dN}{dt} = kN(S - N)$

d. $\frac{dm}{dt} = -km$

35 **[Intermediate]** • **Target: Equations requiring double-angle substitution**

Which differential equation correctly models the statement: "The rate of increase of the number of infected people N is proportional to the product of N and the number of remaining susceptible people $(S - N)$, for fixed population S."?

a. $\frac{dm}{dt} = -km$

b. $\frac{dV}{dt} = kr^2$

c. $\frac{dT}{dt} = -k(T - T_e)$

d. $\frac{dN}{dt} = kN(S - N)$

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36 [Intermediate] • Target: Applications of double-angle formulas

Which differential equation correctly models the statement: "The rate of increase of the volume V of a balloon is proportional to the square of its radius r."?

a. $\frac{dT}{dt} = -k(T - T_e)$

b. $\frac{dm}{dt} = -km$

c. $\frac{dV}{dt} = kr^2$

d. $\frac{dP}{dt} = kP$

37 [Intermediate] • Target: More equation practice

Which differential equation correctly models the statement: "The rate of decrease of a radioactive mass m is proportional to m."?

a. $\frac{dN}{dt} = kN(S - N)$

b. $\frac{dV}{dt} = kr^2$

c. $\frac{dP}{dt} = kP$

d. $\frac{dm}{dt} = -km$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Half-angle formulas for tangent • Cross-Lesson: 9.03 The Binomial Theorem

Which differential equation correctly models the statement: "The rate of increase of a population P is proportional to P itself."?

a. $\frac{dP}{dt} = kP$

b. $\frac{dT}{dt} = -k(T - T_e)$

c. $\frac{dm}{dt} = -km$

d. $\frac{dN}{dt} = kN(S - N)$

39 [Challenging] • Target: Equations with multiple angle types • Cross-Lesson: 1.01 Functions and Function Notation

Find the equation of the tangent line to the circle $x^2 + y^2 = 25$ at $(3, 4)$.

a. $3x + 4y = 25$

b. $x + y = 7$

c. $4x + 3y = 25$

d. $3x - 4y = 25$

40 [Challenging] • Target: Triple-angle preview • Cross-Lesson: 1.02 Domain and Range

Find the area of a sector with radius 4 and central angle $\pi/3$.

a. 2π

b. $\frac{4\pi}{3}$

c. $\frac{8\pi}{3}$

d. 4π

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41 [Challenging] • Target: Graphical verification of a double-angle identity • Cross-Lesson: 1.03 Average Rate of Change

A radioactive substance has half-life 2 years and initial mass 16g. Find the mass remaining after 4 years.

a. 4

b. 16

c. 8

d. 5

42 [Challenging] • Target: Solving equations restricted to a smaller interval • Cross-Lesson: 2.01 Right-Triangle Trigonometry

State the Weierstrass substitution $t = \tan(x/2)$.

a. $\sin x = \frac{2t}{1+t^2}$, $\cos x = \frac{1-t^2}{1+t^2}$, $dx = \frac{2\,dt}{1+t^2}$

b. $x = 2\arctan t$ only

c. $\tan x = 2t$

d. $\sin x = t$, $\cos x = \sqrt{1-t^2}$

43 [Challenging] • Target: Double-angle formulas • Cross-Lesson: 2.02 Radian Measure and Arc Length

A system with more unknowns than equations is called:

a. Consistent

b. Overdetermined

c. Trivial

d. Underdetermined

44 [Challenging] • Target: Solving trigonometric equations • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

Derive the Mollweide's equation: $\frac{a+b}{c} = \frac{\cos\frac{A-B}{2}}{\sin C/2}$ for triangle ABC.

a. Only for right triangles

b. Use the law of cosines

c. Cannot be derived

d. Use the law of sines and angle sum property

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45 [Challenging] • Target: Half-angle formulas • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Find the equation of the tangent to the ellipse $\frac{x^2}{4} + y^2 = 1$ at $(1, 32)$.

a. $\frac{x}{4} - y = 1$

b. $\frac{x}{4} + \frac{\sqrt{3}y}{2} = 1$

c. $x + y = 2$

d. $x + y = 1$

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